Chapter 3: Question 1

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1 Results and Interpretation: Multiple Linear Regression on Advertising Data

In this case ours, a multiple linear regression model was fitted to examine the relationship between sales and advertising expenditures on TV, radio, and newspaper. Table 3.4 presents the estimated coefficients along with their corresponding standard errors, t-statistics, and p-values.

1.1 Null Hypotheses

For each predictor, the reported p-value corresponds to a test of the following null hypotheses:

- **Intercept:** The null hypothesis states that average sales would be zero if there were no spending on TV, radio, or newspaper advertising.
- **TV Advertising:** The null hypothesis states that changes in TV advertising expenditure have no effect on sales.
- **Radio Advertising:** The null hypothesis states that variations in radio advertising spending do not influence sales.
- **Newspaper Advertising:** The null hypothesis states that changes in newspaper advertising spending have no effect on sales, after accounting for TV and radio advertising.

1.2 Interpretation of p-values

The p-value associated with the intercept is less than 0.0001, indicating that average sales are significantly different from zero when no advertising is present. However, this result is mainly of statistical interest and does not have a strong practical implication for marketing strategy.

For TV advertising, the p-value is less than 0.0001, which is far below the conventional significance level of 0.05. Therefore, we reject the null hypothesis and conclude that TV advertising has a statistically significant effect on sales. This suggests that increasing investment in TV advertising is associated with higher sales.

Similarly, the p-value for radio advertising is also less than 0.0001, leading us to reject the null hypothesis. This provides strong evidence that radio advertising significantly contributes to increasing sales, independent of TV and newspaper spending.

However, the p-value for newspaper advertising is 0.8599, which is much greater than 0.05.We fail to reject the null hypothesis for newspaper advertising. This indicates that, after controlling for TV and radio advertising, newspaper spending does not have a statistically significant impact on sales.

1.3 Conclusion

The results suggest that TV and radio advertising are effective channels for increasing sales, as both exhibit strong and statistically significant relationships with sales. In contrast, newspaper advertising does not appear to have a meaningful effect on sales once TV and radio expenditures are taken into account. These findings imply that firms should prioritize investments in TV and radio advertising rather than newspaper advertising when aiming to maximize sales.